



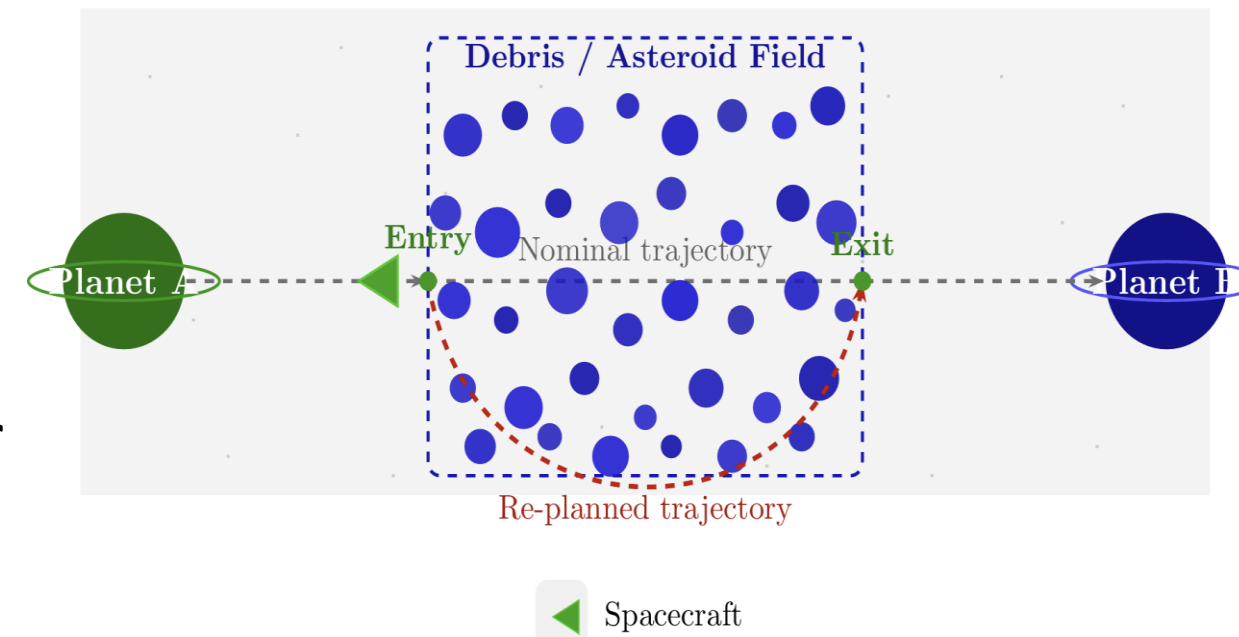
Comparing Controller Schemes for Spacecraft Navigation under Uncertainty

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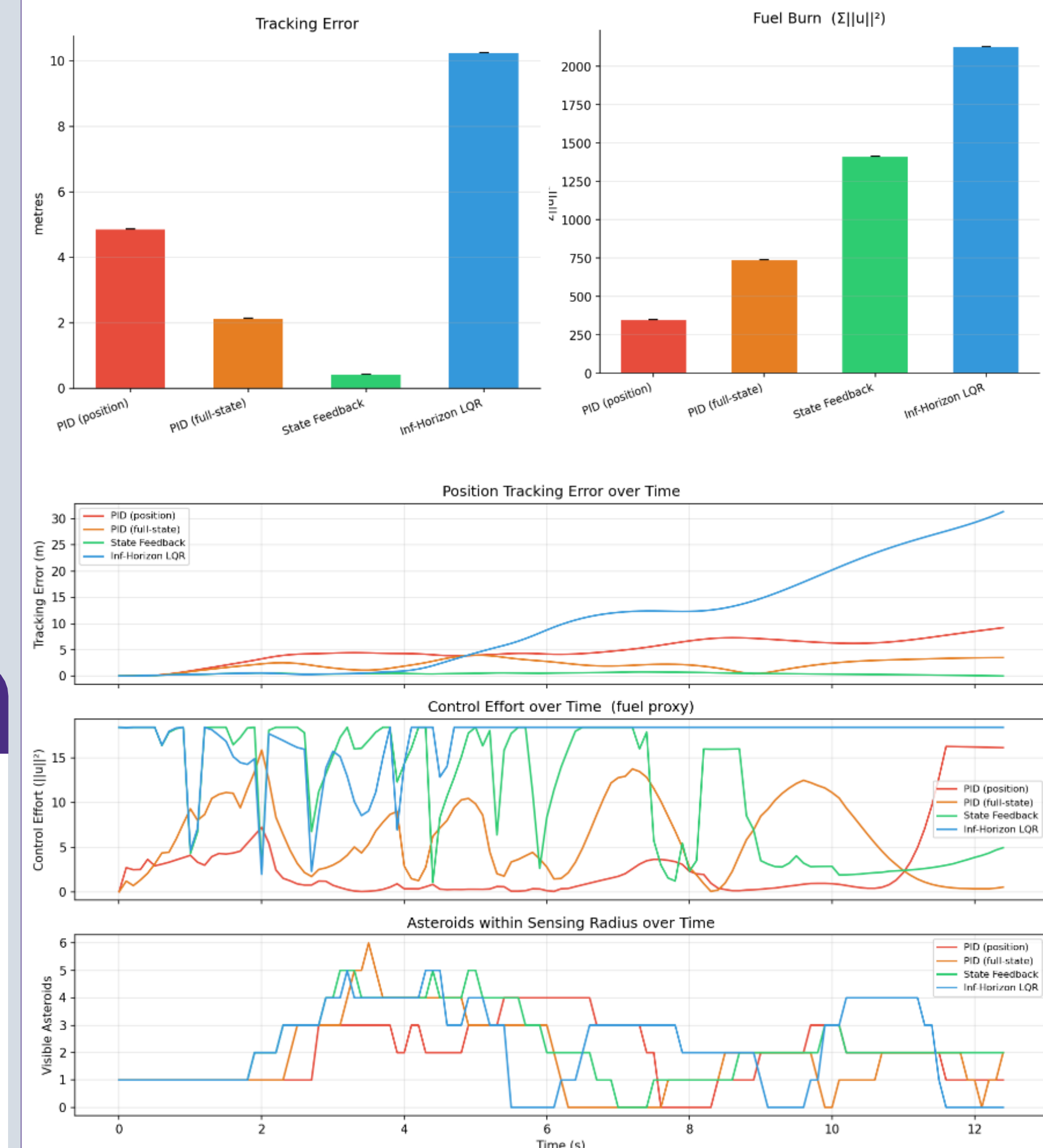
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Abstract

Trajectory optimization is a foundational problem in autonomous systems. Real-world deployments routinely face uncertainty from sensor noise, dynamic obstacles and stochastic disturbances. Deterministic optimization methods produce trajectories that are optimal only under nominal assumptions. This project addresses robust motion planning for an autonomous spacecraft transferring from the orbit of Planet A to Planet B along a precomputed nominal trajectory that is fuel and time-optimal under ideal conditions. Mid-transit, the spacecraft encounters an unanticipated debris or asteroid field. The vehicle enters the region at a fixed initial state and must exit at a specified terminal state to ensure successful orbital insertion.



Results

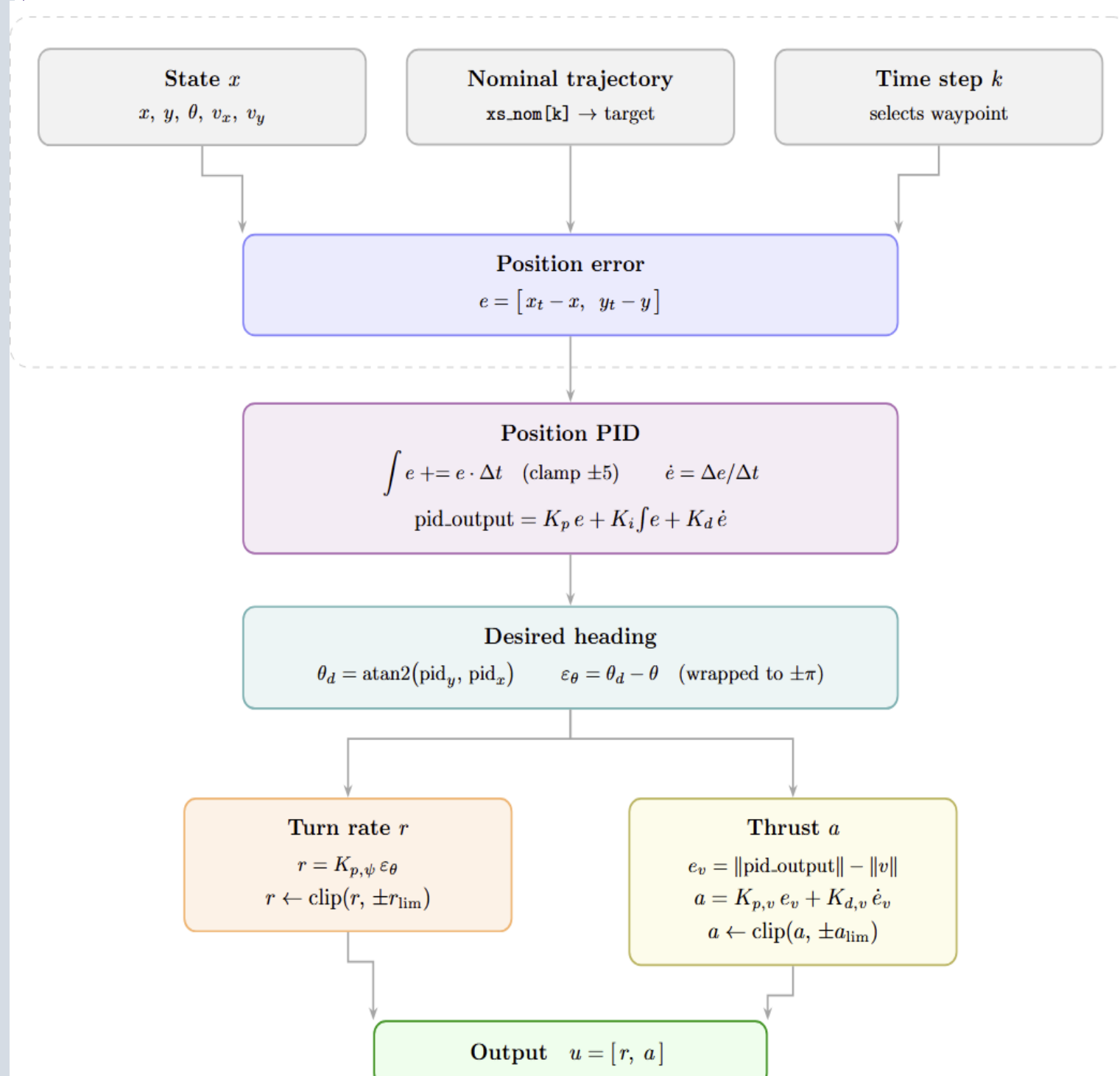


Background and Motivation

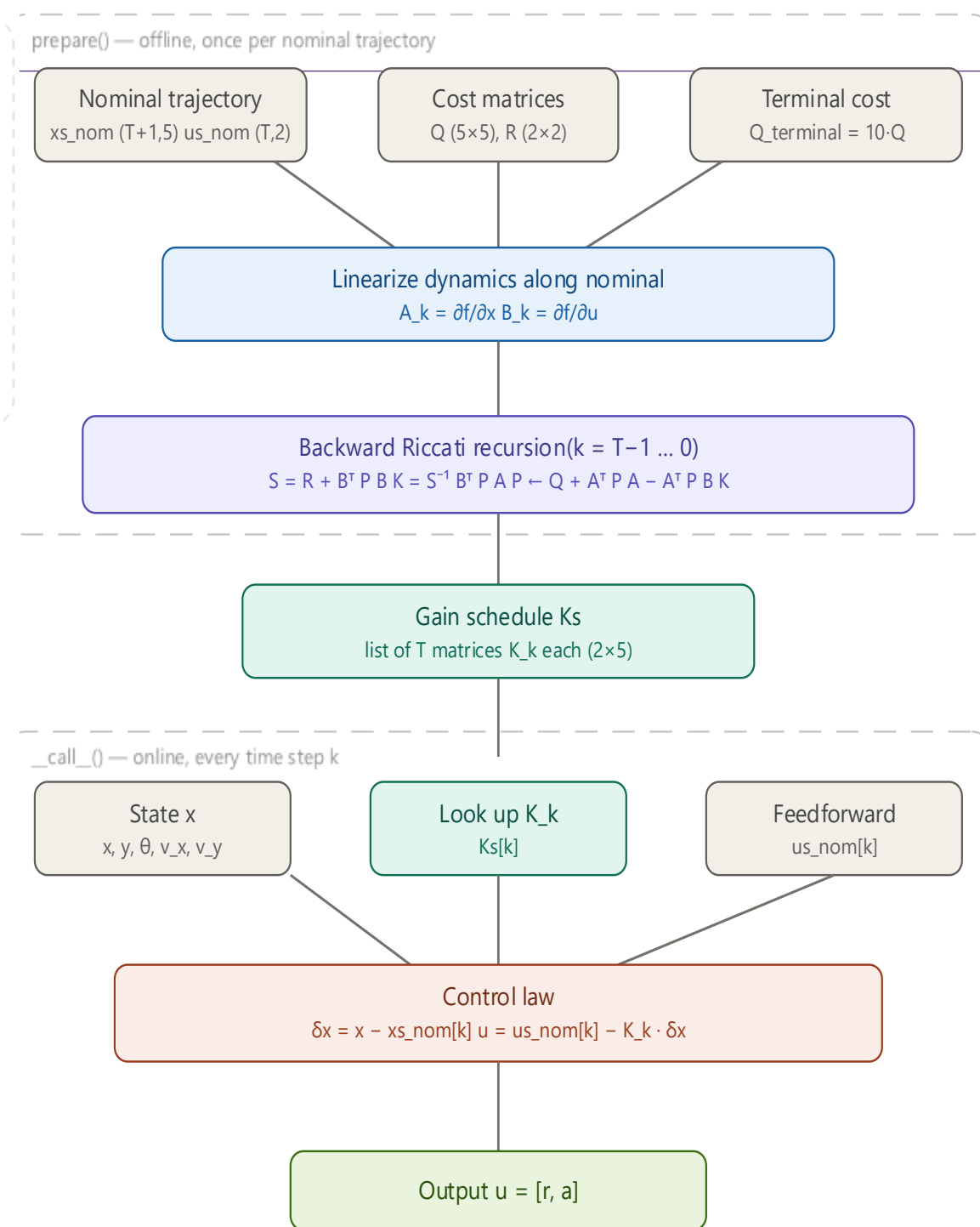
Overview-

- Prior work has explored robust optimization and stochastic control for uncertain dynamical systems in spacecraft guidance and flight trajectory design.
- Limited work exists on systematically comparing controller performance during constrained trajectory adaptation in dynamically changing environments.
- Aim to apply theoretical concepts from previous classes to practical implementation.

Classical (PID)-



State-Feedback (LQR)

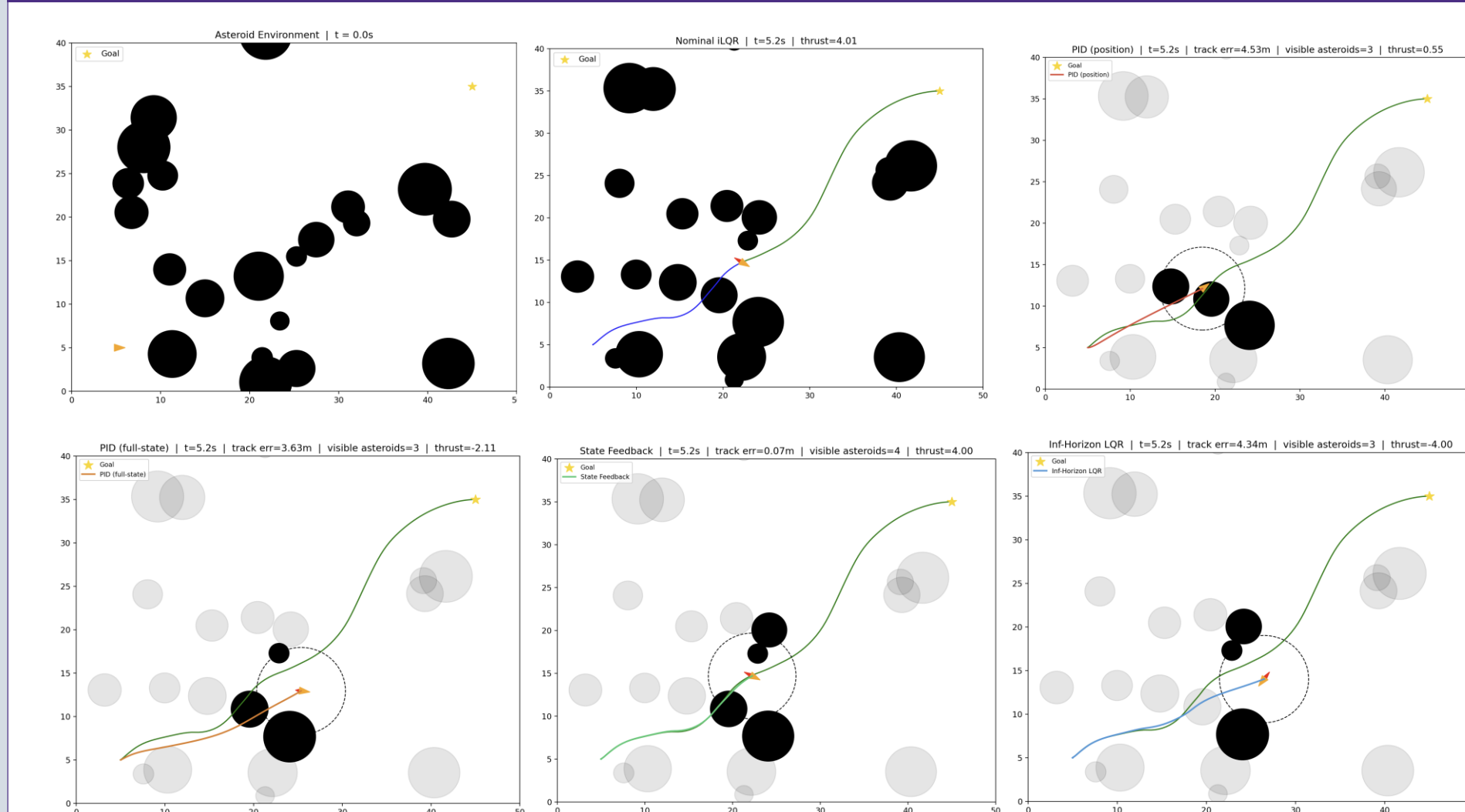


Experiments

Evaluation metrics:

- **Tracking Error:** How much the spacecraft deviates from nominal trajectory.
- **Reaching:** Distance between the goal and the final state
- **Fuel Burn-** How much fuel is burnt while avoiding the obstacles.
- **Time:** computation time required to compute a control input at the current time step.

Integration with Asteroid Environment and Simulation



Conclusions

- Among the tested out controllers State-feedback (LQR) performs the best.
- Future work includes incorporating modern control approaches such as diffusion-guided control and MPPI control.
- Include advanced numerical techniques, such as adaptive-step and symplectic integrators, to improve robustness and long-term stability.
- Account for gravitational effects and orbital dynamics, enabling higher-fidelity simulations of complex physical systems.